
Analog Electronics Lab

Laboratory Manual

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ESE Marks	50
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Corequisites	PBEET304

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Preface

This manual has been prepared to accompany the laboratory course **Analog Electronics Lab (PCEEL308)(2024 scheme)**, offered in Semester 3 of the B.Tech. programme in Electrical and Electronics Engineering. The course is designed to provide hands-on exposure to the design, simulation, and hardware realisation of fundamental analog electronic circuits employing diodes, BJTs, JFETs, MOSFETs, operational amplifiers, and the 555 timer IC.

Each experiment in this manual follows a uniform structure consisting of the aim, theoretical background, pre-lab assignment, components and equipment required, circuit diagram, design procedure, experimental procedure, expected/model graphs, observation tables, results, and viva-voce questions. Students are expected to complete the pre-lab assignment prior to each laboratory session and to record all readings directly in the laboratory record during the conduct of the experiment.

Course Information

Course Objectives

1. To design transistor and operational-amplifier based electronic circuits.
2. To simulate and implement the designed circuits in hardware and validate experimental results against theoretical/simulated predictions.

Course Outcomes

On successful completion of this course, the student will be able to:

CO	Course Outcome	Bloom's Level
CO1	Use the various electronic instruments for conducting experiments.	K3
CO2	Design and develop various electronic circuits using diodes and Zener diodes.	K3
CO3	Design and implement amplifier and oscillator circuits using BJT and JFET.	K3
CO4	Design and implement basic circuits using IC (OPAMP and 555 timer).	K3
CO5	Simulate electronic circuits using circuit simulation software.	K3
CO6	Use PCB layout software for circuit design.	K3

Table 1: Course Outcomes and Bloom's Knowledge Level

K1: Remember, K2: Understand, K3: Apply, K4: Analyse, K5: Evaluate, K6: Create

Scheme of Evaluation

Continuous Internal Evaluation (CIE – 50 Marks)

Component	Marks
Attendance	5
Preparation / Pre-Lab Work, Viva, and Timely Completion of Lab Reports / Record (Continuous Assessment)	25
Internal Examination	20
Total	50

Table 2: CIE Marks Distribution

The continuous assessment component (25 marks) is further broken down as:

- **Preparation and Pre-Lab Work (7 Marks):** Pre-lab assignments/quizzes and understanding of theoretical background.
- **Conduct of Experiments (7 Marks):** Adherence to procedure, accurate execution, safety, skill proficiency, and teamwork.
- **Lab Reports and Record Keeping (6 Marks):** Clarity, completeness, accuracy, and timely submission.
- **Viva Voce (5 Marks):** Ability to explain the experiment, results, and underlying principles.

The final marks for preparation, conduct, viva, and record are computed as the average across all experiments listed in the syllabus.

End Semester Examination (ESE – 50 Marks)

Component	Marks
Procedure / Preliminary Work / Design / Algorithm	10
Conduct of Experiment / Execution of Work / Troubleshooting / Programming	15
Result with Valid Inference / Quality of Output	10
Viva Voce	10
Record	5
Total	50

Table 3: ESE Marks Distribution

- Students shall be permitted to appear for the end-semester examination only upon submission of a duly certified record.
- The external examiner shall endorse the record at the time of the examination.

CO–PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2	3	3	3	3				3	3		
CO3	2	3	3	3	3				3	3		
CO4	2	3	3	3	3				3	3		
CO5	2	3	3	3	3				3	3		
CO6	3								3	3		

Table 4: CO–PO Mapping (1: Slight, 2: Moderate, 3: Substantial, blank: No Correlation)

Text Books and References

1. Robert T. Paynter, *Introductory Electronic Devices and Circuits*, Pearson Education.
2. Boylestad R. L. and Nashelsky L., *Electronic Devices and Circuit Theory*, Pearson Education.
3. Donald A. Neamen, *Electronic Circuits: Analysis and Design*, McGraw Hill.

General Instructions to Students

Laboratory Conduct

1. Students must bring the laboratory record, observation notebook, graph sheets, and a scientific calculator to every session.
2. The pre-lab assignment for the experiment scheduled for the day must be completed and ready for evaluation before entering the laboratory.
3. Students shall enter the laboratory only with permission and shall occupy the workbench allotted to them.
4. Circuit connections must be checked and approved by the faculty-in-charge or laboratory instructor before the power supply is switched on.
5. Readings must be entered directly into the observation notebook at the time of the experiment; fabricated or estimated readings will not be accepted.
6. Calculations, graphs, and inferences must be completed and the experiment got corrected on the same day, wherever possible, or within the stipulated time announced by the instructor.
7. Students must simulate the circuit using the prescribed circuit simulation software prior to hardware implementation, wherever applicable.
8. Mobile phones shall be kept in silent mode and shall not be used during the conduct of the experiment, except as directed for simulation work.
9. Any damage to equipment or components due to negligence shall be reported immediately and may be subject to the institution's equipment-damage policy.
10. Students are responsible for returning all components, probes, and patch cords to their designated locations and for switching off all instruments and the workbench power supply before leaving the laboratory.

Best Practices for Breadboarding

Breadboarding is a fundamental skill in analog electronics. A neat and disciplined approach in breadboarding will help achieving the expected results without any hassle. Following these best practices will help you build reliable circuits and simplify troubleshooting:

- **Plan Your Layout:** Before placing components, mentally (or physically with a drawing) plan the layout of your circuit on the breadboard. Group related components.
- **Shortest Connections:** Use short, neat wires for connections. Long, messy wires can introduce noise, capacitance, and inductance, and make troubleshooting difficult.

- **Color Coding:** Use different colored wires for different purposes (e.g., red for +Vcc, black for ground, blue for signal lines). This greatly aids in tracing connections.
- **Power Rails:** Always connect the power supply rails (+Vcc and GND) to the dedicated power buses on the breadboard.
- **Component Orientation:** Pay close attention to the polarity of components like diodes, LEDs, and electrolytic capacitors.
- **Secure Connections:** Ensure all component leads and wires are firmly inserted into the breadboard holes. Loose connections are a common cause of circuit malfunction.
- **Start Simple:** When troubleshooting, simplify your circuit. Isolate sections and test them individually.
- **Component Identification:** Familiarize yourself with common component packages and their pinouts (e.g., op-amp pin configurations, transistor pinouts).
- **Avoid Overcrowding:** Don't try to cram too many components into a small area. Use a larger breadboard if necessary.
- **Check Continuity:** Before applying power, use a Digital Multimeter (DMM) to check for short circuits between power rails and to verify critical connections.

Observations and Data Recording

Accurate and detailed observations are crucial for a successful lab experience and for writing comprehensive lab reports.

- **Lab Notebook:** Maintain a dedicated lab notebook.
- **Date and Time:** Always start each entry with the date and time.
- **Experiment Title:** Clearly state the experiment title and number.
- **Circuit Diagrams:** Draw clear and labeled circuit diagrams.
- **Component Values:** Record all component values used in your circuit.
- **Measurements:** Tabulate all measured data with appropriate units.
- **Waveforms:** Sketch or capture screenshots of oscilloscope waveforms, labeling axes, amplitudes, and frequencies.
- **Unexpected Results:** Note down any unexpected observations or deviations from expected results. This is often where valuable learning occurs.
- **Troubleshooting Steps:** Document any troubleshooting steps taken and their outcomes.
- **Inferences and Discussions:** Write down your inferences and discuss the implications of your observations.

Lab Report (Record) Submission

1. The first thing to note is that the lab report/record is **NOT A COPY** of the lab manual. The laboratory record is, as its name indicate, a record/report of the experiment done in the lab. The lab manual may include information required for aiding the writing of the report.
2. The lab record must contain, for each experiment: aim, components/equipment list, design, circuit diagram (with all component labels/values clearly shown), experimental procedure, observation tables, model graphs, sample calculations, result, and inferences. Please note that it is not required to write the theory and informational notes that appear in this lab manual.
3. It is suggested to draw the circuit diagrams and observation tables on the left side of the record, and preferably **not on a graph page**. Graph pages are for drawing graphs, waveforms and sample/model curves/graphs.
4. Records must be certified by the faculty-in-charge within one week of completion of the experiment.
5. Submission of a duly certified record is mandatory for eligibility to appear for the end-semester examination.

Evaluation of Pre-Lab Assignments

Two pre-lab assignments are prescribed by the syllabus prior to the regular experiments:

1. Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a cathode-ray oscilloscope (CRO) / digital storage oscilloscope (DSO).
2. Introduction to circuit simulation using a circuit simulation software package (e.g., LTspice, PSpice, Multisim, or NgSpice/QUCS).

Students are expected to complete these foundational exercises before attempting the numbered experiments listed in Table 6.

Safety Guidelines

Safety is paramount in any laboratory environment. Please read and adhere to the following safety guidelines:

Safety Note

1. **No Food or Drink:** Do not bring or consume food or drinks in the lab.
2. **Work Area:** Keep your workspace clean and organised. Cluttered benches can lead to accidents.
3. **Before Power On:** Always verify the power supply voltage and polarity before connecting it to the circuit. Reversed polarity can permanently damage diodes, transistors, op-amps, and ICs.
4. **Power Off First:** Switch off the power supply before making, modifying, or removing any circuit connections.
5. **Component Handling:** Handle components carefully. Be aware of the polarity of diodes, LEDs, and electrolytic capacitors. Do not exceed the maximum voltage, current, or power ratings of any active or passive component. Always compute and verify current-limiting resistor values before energising a circuit. Use current-limiting resistors in series with LEDs, diodes, and Zener diodes during testing to prevent excessive current flow. Electrolytic capacitors are polarised. Verify polarity before connecting; reversed polarity may cause the capacitor to overheat, vent, or rupture.
6. **Tool Usage:** Use tools (wire strippers, pliers, screwdrivers) correctly and for their intended purpose.
7. **No Short-circuits:** Avoid short-circuiting the output terminals of a regulated DC power supply or the output of an operational amplifier.
8. **Hot Components:** Be aware of the heat-sink and case temperature of power transistors and voltage regulators during prolonged operation; avoid direct contact.
9. Do not work on live circuits with wet hands or in a wet environment.
10. **Report Accidents:** In the event of a component overheating, emitting smoke, or unusual odour, immediately switch off the power supply and inform the laboratory instructor before attempting further investigation. Immediately report any accidents, spills, or equipment malfunctions to the lab instructor or faculty-in-charge or TA.
11. **No Horseplay:** Maintain a professional demeanour and avoid any behaviour that could endanger yourself or others.

12. **Emergency:** Be aware of the location of emergency exits and first-aid kits. Familiarise yourself with the location of the mains isolation switch and the fire extinguisher in the laboratory.
13. **Before Leaving:** Ensure your workstation is clean and all equipment is turned off before leaving the lab.

Precautions

1. Identify the pin configuration (pinout) of every IC and transistor from its datasheet before inserting it into the breadboard.
2. Avoid loose or floating connections, particularly at the input terminals of operational amplifiers, since floating CMOS/op-amp inputs may result in unpredictable circuit behaviour.
3. Keep signal leads and connecting wires as short as practicable to minimise stray capacitance and inductance, particularly in high-frequency oscillator circuits.
4. Re-verify all connections against the circuit diagram before switching on the power supply, and recheck after every modification.
5. Handle CRO/DSO probes with care; ensure the ground (common) lead of the probe is connected to the circuit ground/common point to avoid ground loops and inaccurate measurements.

List of Equipment and Software

Introduction to Essential Lab Equipment

Familiarity with the following equipment is essential for performing experiments in this lab.

Digital Storage Oscilloscope (DSO)

- **Purpose:** The DSO is used to visualize and analyze time-varying electrical signals (waveforms). It displays voltage as a function of time.
- **Key Controls:**
 - **Vertical (Volts/Div):** Controls the voltage scale on the screen (voltage per vertical division).
 - **Horizontal (Sec/Div):** Controls the time scale on the screen (time per horizontal division).
 - **Trigger:** Stabilizes repetitive waveforms on the screen. Adjust trigger level and mode (edge, auto, normal).
 - **Channels:** DSOs typically have multiple input channels (CH1, CH2, etc.) to view multiple signals simultaneously.
 - **Cursors/Measurements:** Use cursors to precisely measure voltage and time intervals. Most DSOs have built-in automated measurement functions (e.g., V_{pp} , V_{rms} , frequency, period).
- **Usage Tips:**
 - Always use a proper probe (1X or 10X). Understand the attenuation factor. Calibrate probes if necessary.
 - Ground connections are crucial for accurate measurements.
 - Start with auto-set if you're unsure of the signal characteristics, then fine-tune.

Function Generator

- **Purpose:** Generates various standard waveforms (sine, square, triangle, pulse) at specified frequencies and amplitudes.
- **Key Controls:**
 - **Waveform Selection:** Selects the desired output waveform.
 - **Frequency:** Sets the output frequency.
 - **Amplitude:** Sets the peak-to-peak voltage of the output waveform.
 - **DC Offset:** Adds a DC voltage component to the output signal.
 - **Output Impedance:** Typically 50 Ω . Be aware of impedance matching.

Usage Tips:

- Connect the output of the function generator to your circuit's input.
- Always check the output on the oscilloscope before connecting to your circuit to ensure it's set correctly.

DC Power Supply

- **Purpose:** Provides stable DC voltage and current to power your circuits.
- **Key Controls:**
 - **Voltage Adjustment:** Sets the output voltage.
 - **Current Limit:**¹ Sets/limits the maximum current the supply can deliver. This is a crucial safety feature to protect your circuit from excessive current.
 - **Output On/Off:** Turns the power supply output on or off.

Usage Tips:

- Always set the voltage and current limit before connecting to your circuit. Start with a lower voltage and gradually increase if needed.
- If the current limit indicator lights up, it means your circuit is drawing more current than allowed, possibly indicating a short circuit or fault. Turn off power immediately and troubleshoot.

Digital Multimeter (DMM)

- **Purpose:** Measures various electrical parameters like voltage (AC/DC), current (AC/DC), resistance, and continuity.
- **Key Functions:**
 - **Voltage Measurement (V):** Connect in parallel with the component or circuit portion you want to measure. Select AC/DC option appropriately.
 - **Current Measurement (A):** Connect in series with the circuit. This requires breaking the circuit. Be cautious about the current range.
 - **Resistance Measurement (Ω):** Measure resistance of disconnected components. Never measure resistance in an active circuit.
 - **Continuity Test:** Beeps if there is a low-resistance path (useful for checking connections).

Usage Tips:

- Always select the correct function (V, A, Ω) and range before making a measurement.
- For current measurements, always connect the DMM in series. Incorrect connection can blow the internal fuse.
- Ensure the circuit is de-energized when measuring resistance.

¹Note that by adjusting this control, one cannot vary the current. The current drawn from a voltage-source power supply is decided by the load resistance.

Equipment / Software / Components	Typical Use in this Lab
Regulated DC Power Supply (dual/triple output)	Biasing of transistor, op-amp, and 555 timer circuits
Function / Signal Generator	Input excitation for amplifiers, filters, and frequency-response studies
Digital Storage Oscilloscope (DSO)	Measurement of waveforms, amplitude, frequency, and phase shift
Digital Multimeter (DMM)	DC bias point and resistance/voltage/current measurements
Breadboard and connecting wires	Circuit prototyping
Active and Passive Components (diodes, Zener diodes, BJTs, JFETs, MOSFETs, resistors, capacitors, inductors)	Circuit construction across experiments
Operational Amplifier ICs (e.g., $\mu A741$, LM358)	Op-amp based experiments
Comparator IC (e.g., LM311)	Comparator and Schmitt trigger experiments
555 Timer IC	Astable and monostable multivibrator experiments
Circuit Simulation Software (e.g., LTspice / Qspice / NgSpice-QUCS)	Pre-implementation simulation of all designed circuits
PCB Layout Software (e.g., KiCad)	Introduction to PCB design

Table 5: Indicative list of equipment and software used in the course

List of Experiments

Table 6: List of Experiments – Analog Electronics Lab (PCEEL308)

Expt. No.	Title	CO
PL-1	Measurement of current, voltage, frequency, and phase shift of a signal in an RC network using a CRO/DSO (Pre-Lab)	CO1
PL-2	Introduction to circuit simulation using a circuit simulation software (Pre-Lab)	CO5
1	Clipping and clamping circuits using diodes	CO2
2	Basic RC circuits – High-pass and Low-pass filters	CO1
3	RC-coupled amplifier using BJT in CE configuration – Measurement of gain, bandwidth, and plotting of frequency response	CO3
4	Emitter follower amplifier	CO3
5	JFET amplifier – Measurement of gain, bandwidth, and plotting of frequency response	CO3
6	MOSFET amplifier	CO3
7	Design and testing of voltage regulators – Zener and series	CO2
8	Design and setup of inverting and non-inverting amplifiers	CO4
9	Op-amp circuits – Scale changer, adder, integrator, and differentiator	CO4
10	Precision rectifier using op-amp	CO4
11	Op-amp oscillators – RC phase-shift and Wien bridge oscillator	CO4
12	Op-amp oscillator – LC oscillator (Colpitts or Hartley)	CO4
13	Waveform generation – Square, triangular, and sawtooth waveform generation using op-amps	CO4
14	Basic comparator and Schmitt trigger circuits using op-amp (using comparator ICs such as LM311)	CO4
15	Active filters (one each of high-pass and low-pass)	CO4
16	Instrumentation amplifier	CO4
17	Astable and monostable multivibrator circuits using 555 timer IC	CO4
18	Introduction to PCB layout software	CO6

Format Followed for Each Experiment

Each experiment in the chapters that follow is presented under the following uniform headings:

1. **Aim**
2. **Pre-Lab Assignment**

3. **Components and Equipment Required**
4. **Theory**
5. **Circuit Diagram**
6. **Design**
7. **Procedure**
8. **Precautions**
9. **Model Graph / Expected Waveform**
10. **Observation Table**
11. **Result**
12. **Questions**

The students are expected to submit the report of each experiment, containing the following:

1. **Aim**
2. **Pre-Lab Assignment**
3. **Components and Equipment Required**
4. **Circuit Diagram**
5. **Design**
6. **Procedure**
7. **Model Graph / Expected Waveform**
8. **Observation Tables**
9. **Results/Conclusions**

Experiment 1

Op-Amp Circuits-II: Adder and Subtractor

1.1 Aim

To design, construct, and experimentally verify the performance of Operational Amplifier (Op-Amp) mathematical circuits using IC 741. The specific objectives are:

1. To implement an Inverting Summing Amplifier (Adder) and verify that $V_{out} = -(V_1 + V_2)$.
2. To implement a Difference Amplifier (Subtractor) and verify that $V_{out} = V_2 - V_1$.
3. To observe and record the input and output waveforms for both DC and AC input signals.

1.2 Pre-Lab Assignment

Before performing the experiment, students must complete the following analytical tasks:

1. Using the virtual ground concept, derive the expression for the output voltage of an inverting summing amplifier with three inputs (V_1, V_2, V_3) and input resistors (R_1, R_2, R_3) with a feedback resistor R_f .
2. Derive the output voltage equation for a difference amplifier circuit when $R_1 \neq R_2 \neq R_3 \neq R_4$. Show how it simplifies to $V_{out} = \frac{R_2}{R_1}(V_2 - V_1)$ when $R_3 = R_1$ and $R_4 = R_2$.
3. Define Common-Mode Rejection Ratio (CMRR) and explain why matched resistor tolerances are critical for obtaining high CMRR in a subtractor circuit.
4. Calculate the expected DC output voltage for an inverting adder ($R_1 = R_2 = R_f = 10 \text{ k}\Omega$) when $V_1 = +2.0 \text{ V}$ and $V_2 = -1.5 \text{ V}$.

1.3 Components and Equipment Required

The components and test instruments required for this experiment are listed in Table 1.1.

1.4 Theory

Operational amplifiers can perform linear mathematical operations directly on analog voltage signals. Negative feedback forces the differential input voltage to near zero, simplifying circuit analysis through the virtual short/ground approximation.

Table 1.1: Components and Equipment Specifications

S. No.	Item Name	Specification / Range	Quantity
1	Operational Amplifier IC	IC 741 (8-pin DIP)	1 No.
2	Resistors	10 k Ω (Precision 1%), 3.3 k Ω	4 Nos., 1 No.
3	Dual DC Power Supply	± 15 V fixed/variable	1 No.
4	Auxiliary DC Source	0 – 5 V variable	1 No.
5	Function Generator	1 Hz – 1 MHz, Dual/Sinusoidal output	1 No.
6	Digital Storage Oscilloscope (DSO)	Dual channel, 20 MHz or higher	1 No.
7	Digital Multimeter (DMM)	4 $\frac{1}{2}$ digit display	1 No.
8	Breadboard & Connecting Wires	Standard multi-contact, single-strand wires	As required

1.4.1 Inverting Summing Amplifier (Adder)

In an inverting summing configuration, multiple input voltages are connected through individual resistors to the inverting input terminal (pin 2) of the op-amp. The non-inverting terminal (pin 3) is connected to ground through an offset-minimizing resistor R_{comp} .

Because the inverting node is maintained at virtual ground (0 V), Kirchhoff's Current Law (KCL) at the summing junction gives:

$$I_1 + I_2 = I_f \implies \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} = \frac{0 - V_{out}}{R_f} \quad (1.1)$$

Rearranging for V_{out} :

$$V_{out} = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 \right) \quad (1.2)$$

When all resistors are equal ($R_1 = R_2 = R_f = R$):

$$V_{out} = -(V_1 + V_2) \quad (1.3)$$

The negative sign indicates an overall 180° inversion of the sum.

1.4.2 Difference Amplifier (Subtractor)

A difference amplifier amplifies the potential difference between two input signals while rejecting any common-mode voltage. V_1 is fed to the inverting input via R_1 , and V_2 is fed to the non-inverting input via a voltage divider formed by R_3 and R_4 .

Applying the superposition principle:

1. With $V_2 = 0$ V, the circuit acts as an inverting amplifier: $V_{out1} = -\frac{R_2}{R_1} V_1$.
2. With $V_1 = 0$ V, the voltage at the non-inverting pin is $V_+ = V_2 \left(\frac{R_4}{R_3 + R_4} \right)$. The circuit acts as a non-inverting amplifier:

$$V_{out2} = V_+ \left(1 + \frac{R_2}{R_1} \right) = V_2 \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{R_1 + R_2}{R_1} \right) \quad (1.4)$$

Combining both contributions yields the net output:

$$V_{out} = V_{out1} + V_{out2} = V_2 \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{R_1 + R_2}{R_1} \right) - \frac{R_2}{R_1} V_1 \quad (1.5)$$

If $R_1 = R_3$ and $R_2 = R_4$:

$$V_{out} = \frac{R_2}{R_1} (V_2 - V_1) \quad (1.6)$$

For unity differential gain ($R_1 = R_2 = R_3 = R_4 = R$):

$$V_{out} = V_2 - V_1 \quad (1.7)$$

1.5 Circuit Diagram

The schematics for the Inverting Adder and Subtractor are shown in Figures 1.1 and 1.2.

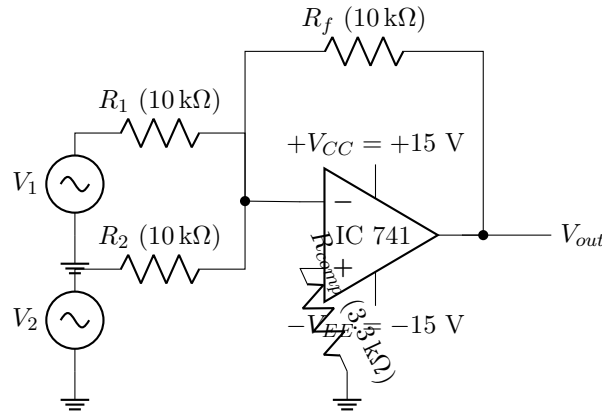


Figure 1.1: Inverting Summing Amplifier (Adder) Circuit Diagram

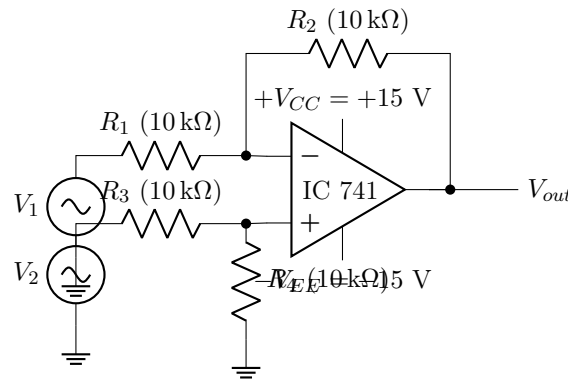


Figure 1.2: Difference Amplifier (Subtractor) Circuit Diagram

1.6 Design

1.6.1 Adder Design

1. Select $R_1 = R_2 = R_f = 10 \text{ k}\Omega$ for unity gain scaling.
2. Calculate R_{comp} to minimize output offset voltage due to input bias currents:

$$R_{comp} = R_1 \parallel R_2 \parallel R_f = \frac{10 \text{ k}\Omega}{3} \approx 3.33 \text{ k}\Omega \quad (\text{Select standard value: } 3.3 \text{ k}\Omega) \quad (1.8)$$

1.6.2 Subtractor Design

1. Select $R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$ using matched 1% tolerance metal film resistors.
2. Verify that $R_1 \parallel R_2 = R_3 \parallel R_4 = 5 \text{ k}\Omega$, ensuring inherent resistance symmetry at both op-amp input terminals.

1.7 Procedure

1.7.1 Adder Circuit Setup

1. Wire the IC 741 on the breadboard according to Figure 1.1. Connect $\pm 15 \text{ V}$ power supplies.

2. **DC Analysis:** Apply DC voltages from auxiliary supplies to V_1 and V_2 as listed in Table 1.2. Measure V_{out} with a digital multimeter and compare with $-(V_1 + V_2)$.
3. **AC Analysis:** Apply sinusoidal inputs at 1 kHz: $V_1 = 1.0 V_{p-p}$ and $V_2 = 1.5 V_{p-p}$ from synchronized generator outputs. Observe V_{out} on the DSO and confirm the sum and 180° phase relation.

1.7.2 Subtractor Circuit Setup

1. Reconfigure the circuit according to Figure 1.2.
2. **DC Analysis:** Apply DC voltages to V_1 and V_2 as listed in Table 1.3. Measure V_{out} using a multimeter and verify $V_{out} = V_2 - V_1$.
3. **AC Analysis:** Apply two sinusoidal signals at 1 kHz with unequal amplitudes ($V_1 = 1.0 V_{p-p}$, $V_2 = 2.5 V_{p-p}$) in phase. Measure V_{out} on the DSO and verify that $V_{out} = 1.5 V_{p-p}$.

1.8 Precautions

1. Turn off power before making any circuit modifications.
2. Ensure that the algebraic sum or difference of the input voltages multiplied by the stage gain does not cause V_{out} to exceed the op-amp saturation limits ($\approx \pm 13 V$).
3. Use matched resistors (1% tolerance) in the subtractor circuit to maintain high Common-Mode Rejection Ratio (CMRR).

1.9 Model Graph / Expected Waveforms

Figure 1.3 illustrates the expected AC signal relationships at 1 kHz for both operational modes.

Expected Inverting Adder Waveforms ($V_1 = 1.0 V$, $V_2 = 1.5 V$)

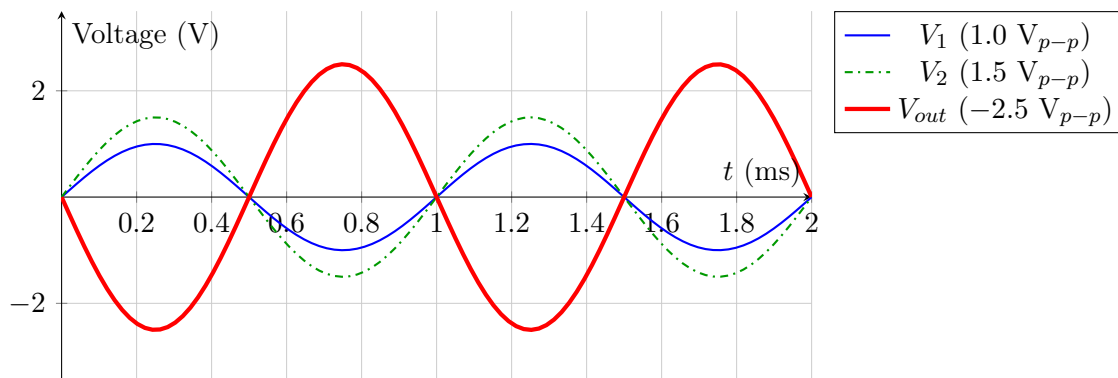


Figure 1.3: Time-Domain Waveforms for Inverting Summing Amplifier

1.10 Observation Tables

1.10.1 Adder Circuit DC Measurements

Record the DC summation observations in Table 1.2.

Table 1.2: Inverting Summing Amplifier DC Verification

S. No.	V_1 (V)	V_2 (V)	Theoretical V_{out} (V)	Practical V_{out} (V)	% Error
1	+1.0	+2.0	-3.0		
2	+2.5	-1.0	-1.5		
3	-1.5	-2.0	+3.5		
4	+0.5	+0.5	-1.0		

1.10.2 Subtractor Circuit DC Measurements

Record the DC difference observations in Table 1.3.

Table 1.3: Difference Amplifier DC Verification

S. No.	V_1 (V)	V_2 (V)	Theoretical V_{out} (V)	Practical V_{out} (V)	% Error
1	+1.0	+3.0	+2.0		
2	+2.5	+1.0	-1.5		
3	-1.0	+2.0	+3.0		
4	+2.0	+2.0	0.0		

1.11 Result

The performance of Op-Amp Adder and Subtractor circuits using IC 741 was successfully evaluated:

- The inverting summing amplifier was verified for DC and AC inputs, confirming that $V_{out} = -(V_1 + V_2)$ with an average experimental deviation of _____%.
- The difference amplifier was verified, confirming $V_{out} = V_2 - V_1$ with an average experimental deviation of _____%.

1.12 Questions

1. How can an inverting summing amplifier be modified to compute the mathematical average of three input signals ($V_{out} = -\frac{V_1+V_2+V_3}{3}$)?
2. What happens to the common-mode rejection capability of a subtractor circuit if one of the resistors has a 5% tolerance mismatch?
3. How does an Instrumentation Amplifier overcome the low input impedance limitations of a basic single-op-amp subtractor?
4. If $V_1 = 5\text{ V}$ and $V_2 = 10\text{ V}$ are applied to an adder configured with unity gain powered at $\pm 15\text{ V}$, will the output be -15 V ? Explain why or why not.
5. Describe how a non-inverting summing amplifier can be designed using a single op-amp.

““

Experiment 2

Op-Amp Circuits-II: Adder and Subtractor

2.1 Aim

To design, assemble, and test a summing amplifier (adder) and a difference amplifier (subtractor) using the IC 741 operational amplifier, and to verify their input–output relationships for both DC and AC input signals.

2.2 Pre-Lab Assignment

- (i) Review the virtual-short and zero-input-current assumptions of the ideal op-amp (introduced in Experiment ??) and their application to circuits with multiple input signals.
- (ii) For a summing amplifier with two/three inputs and a target output relationship assigned to your batch (e.g., $v_{out} = -(v_1 + 2v_2)$), design the required resistor values.
- (iii) For a difference amplifier with a target gain assigned to your batch, design the required resistor values, ensuring the condition for common-mode rejection (Equation (2.6)) is satisfied.
- (iv) Simulate both designed circuits in LTspice (Pre-Lab 2) with appropriate DC and/or sinusoidal sources at each input, and verify the expected output algebraically combines the inputs as designed.

2.3 Components and Equipment Required

S.No.	Component/Equipment	Specification
1	Operational Amplifier IC	$\mu A741$ (or equivalent), 1 no.
2	Resistors	R_1, R_2, R_3, R_f as per design
3	Dual Regulated DC Power Supply	$\pm 15\text{ V}$
4	Function Generator(s)	Sine wave, 1 Hz – 1 MHz (2 nos., if available)
5	Additional DC Power Supply	0–5 V (for DC input levels)
6	Digital Storage Oscilloscope	2-channel, 20–50 MHz bandwidth
7	Digital Multimeter	For DC voltage measurements
8	Breadboard and connecting wires	–

Table 2.1: Components and equipment list.

2.4 Theory

Building on the inverting and non-inverting amplifier configurations of Experiment ??, the same ideal op-amp assumptions – zero input current and the virtual-short condition – allow the construction of circuits that perform linear arithmetic operations (weighted addition and subtraction) directly on input voltages.

2.4.1 Summing Amplifier (Inverting Adder)

Figure 2.1 shows a summing amplifier with three input voltages v_1 , v_2 , v_3 applied through resistors R_1 , R_2 , R_3 to the inverting input, which is held at virtual ground. Since no current enters the op-amp's inverting terminal, Kirchhoff's Current Law at this node gives

$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = -\frac{v_{out}}{R_f}, \quad (2.1)$$

so that

$$v_{out} = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right). \quad (2.2)$$

Each input is thus independently weighted by the ratio R_f/R_i , and the outputs are summed and inverted. Choosing $R_1 = R_2 = R_3 = R_f$ realises a simple (unity-weighted) inverting adder, $v_{out} = -(v_1 + v_2 + v_3)$.

2.4.2 Difference Amplifier (Subtractor)

Figure 2.2 shows the standard single-op-amp difference amplifier. Input v_1 drives the inverting side through R_1 , with feedback resistor R_f ; input v_2 drives the non-inverting side through a resistor R_2 , with R_3 from the non-inverting terminal to ground. By superposition (or direct node analysis using the virtual-short condition):

The non-inverting input voltage is set by the divider formed by R_2 and R_3 :

$$v_+ = v_2 \cdot \frac{R_3}{R_2 + R_3}. \quad (2.3)$$

Since $v_- = v_+$ (virtual short) and applying KCL at the inverting node:

$$\frac{v_1 - v_-}{R_1} = \frac{v_- - v_{out}}{R_f}. \quad (2.4)$$

Solving these together gives the general result

$$v_{out} = \frac{R_f}{R_1} \left(1 + \frac{R_1}{R_f}\right) \left(\frac{R_3}{R_2 + R_3}\right) v_2 - \frac{R_f}{R_1} v_1. \quad (2.5)$$

If the resistors are chosen in the matched ratio

$$\frac{R_3}{R_2} = \frac{R_f}{R_1}, \quad (2.6)$$

(a condition that also maximises the common-mode rejection ratio, CMRR, of the practical circuit), the expression simplifies to the ideal difference amplifier relationship

$$v_{out} = \frac{R_f}{R_1} (v_2 - v_1). \quad (2.7)$$

With $R_1 = R_2$ and $R_f = R_3$, Equation (2.7) shows the circuit produces an output proportional to the *difference* of the two input signals, rejecting any signal common to both inputs.

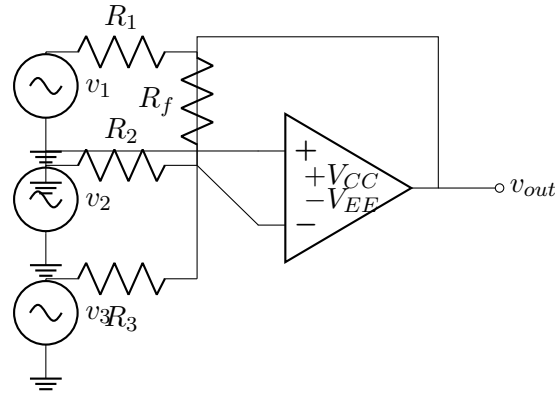


Figure 2.1: Three-input inverting summing amplifier (adder). All inputs sum at the virtual-ground node and are weighted by R_f/R_i .

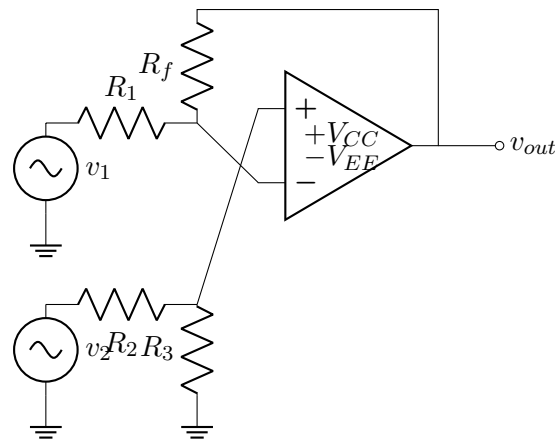


Figure 2.2: Difference amplifier (subtractor). With $R_1 = R_2$ and $R_f = R_3$, $v_{out} = (R_f/R_1)(v_2 - v_1)$.

2.5 Circuit Diagram

2.6 Design

2.6.1 Summing Amplifier

Given target weights k_1, k_2, k_3 for a desired relationship $v_{out} = -(k_1 v_1 + k_2 v_2 + k_3 v_3)$:

- Choose a convenient feedback resistor R_f (typically 10–100 k Ω).
- Calculate $R_i = R_f/k_i$ for each input i , from Equation (2.2).
- Round to the nearest standard resistor values and recompute the actual weights realised.
- Ensure the maximum expected $|v_{out}|$ (for the largest anticipated combination of inputs) remains within the op-amp's output voltage swing (typically ± 13 V for ± 15 V supplies).

2.6.2 Difference Amplifier

Given a target gain A_d for $v_{out} = A_d(v_2 - v_1)$:

- Choose $R_1 = R_2$ (typically 10 k Ω) for a symmetric design.
- Calculate $R_f = R_3 = A_d R_1$, satisfying the matched-ratio condition of Equation (2.6).

- (c) Use close-tolerance (e.g., 1%) resistors where possible, since mismatch in the ratio R_3/R_2 vs. R_f/R_1 directly degrades the common-mode rejection ratio of the practical circuit.
- (d) Verify the maximum expected $|v_{out}|$ remains within the op-amp's output swing.

2.7 Procedure

2.7.1 Summing Amplifier

1. Connect the $\pm 15\text{ V}$ supply to the op-amp and assemble the adder circuit of Figure 2.1 using the designed resistor values.
2. **DC test:** Apply known DC voltages (using a DC supply and/or potential dividers) to v_1 , v_2 , v_3 in turn (with the others grounded), and measure v_{out} with a multimeter/DSO for each case. Verify against Equation (2.2).
3. Apply all three DC inputs simultaneously and verify v_{out} equals the algebraic sum predicted by Equation (2.2).
4. **AC test:** Apply a sine wave (e.g., 1 kHz, small amplitude) to v_1 with v_2, v_3 grounded; observe v_{out} on the DSO and verify the amplitude and phase (inverted) match theory.
5. Repeat with two sine inputs of different frequencies (if two function generators are available) and observe the summed waveform at the output.

2.7.2 Difference Amplifier

1. Assemble the subtractor circuit of Figure 2.2 using the designed resistor values.
2. **DC test:** Apply known DC voltages to v_1 and v_2 (individually and together) and measure v_{out} ; verify against Equation (2.7).
3. **Common-mode test:** Apply equal DC voltages to both v_1 and v_2 (i.e., $v_1 = v_2$) and verify that $v_{out} \approx 0$, demonstrating common-mode rejection.
4. **AC test:** Apply a sine wave to v_1 with v_2 grounded (and vice versa); observe and record v_{out} on the DSO in each case.
5. If two function generators are available, apply sine waves of the same frequency and amplitude but different phase (or one shifted by a DC offset) to v_1 and v_2 , and verify v_{out} corresponds to their difference.

2.8 Precautions

- Connect the op-amp supply pins with correct polarity before applying any input signal.
- Keep all input voltage levels and their weighted combination within the linear (non-saturating) output range of the op-amp.
- Use close-tolerance resistors for the difference amplifier to obtain good common-mode rejection; measure actual resistor values with a multimeter if precision matters.
- Ensure a common ground between all DC/AC sources, the circuit, and the DSO.
- Switch off the power supply before changing any circuit connections.

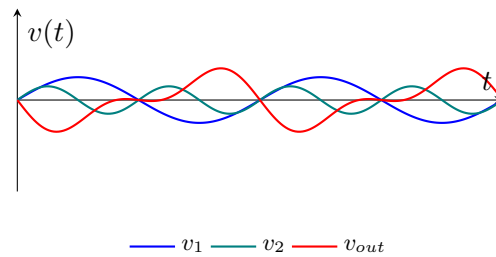


Figure 2.3: Expected waveforms for the summing amplifier with two sinusoidal inputs of different frequency ($R_1 = R_2 = R_f$): the output is the inverted sum of the two inputs.

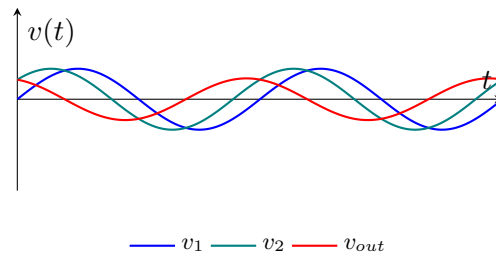


Figure 2.4: Expected waveforms for the difference amplifier with two equal-amplitude, equal-frequency inputs of different phase ($R_1 = R_2$, $R_f = R_3$): the output amplitude reflects only the phase/amplitude difference between v_1 and v_2 .

2.9 Model Graph / Expected Waveforms

2.10 Observation Table

v_1 (V)	v_2 (V)	v_3 (V)	v_{out} (V)
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Table 2.2: Summing amplifier: DC observation table (measured vs. Equation (2.2)).

v_1 (V)	v_2 (V)	v_{out} (theory)	v_{out} (measured)	% Error
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Table 2.3: Difference amplifier: DC observation table (including the equal-input common-mode test).

2.11 Result

Summing and difference amplifier circuits were designed, assembled using the IC 741, and tested with both DC and AC input signals. The measured output in each case was found to be in close

agreement with the theoretical relationships of Equations (2.2) and (2.7), and the difference amplifier was verified to substantially reject common-mode input signals when the resistor ratios satisfied Equation (2.6).

2.12 Questions

- Q1. Using the virtual-ground concept, derive Equation (2.2) for the three-input summing amplifier from first principles.
- Q2. How would you modify the summing amplifier circuit to perform *subtraction* of two signals using only inverting-adder and inverting-amplifier (unity gain) stages, without using the single-op-amp difference amplifier of Figure 2.2?
- Q3. Why is the matched-ratio condition of Equation (2.6) important for common-mode rejection in the difference amplifier? What happens to the CMRR if $R_3/R_2 \neq R_f/R_1$?
- Q4. Define Common-Mode Rejection Ratio (CMRR) and explain why a high CMRR is desirable in instrumentation applications.
- Q5. What is a key limitation of the single-op-amp difference amplifier of Figure 2.2 regarding input impedance, particularly when R_1 is small? How does an instrumentation amplifier (using three op-amps) overcome this limitation?
- Q6. If all three inputs of the summing amplifier in Figure 2.1 are driven with the same DC voltage V , and $R_1 = R_2 = R_3 = R_f$, what is v_{out} ? Verify this against your observation table.
- Q7. Explain, with an example, one practical application each of a summing amplifier and a difference amplifier in analogue signal processing systems.

Appendix A

Manufacturer Datasheets



2N7000

N-Channel Enhancement-Mode Vertical DMOS FET

Features

- Free from Secondary Breakdown
- Low Power Drive Requirement
- Ease of Paralleling
- Low C_{ISS} and Fast Switching Speeds
- Excellent Thermal Stability
- Integral Source-Drain Diode
- High Input Impedance and High Gain

Applications

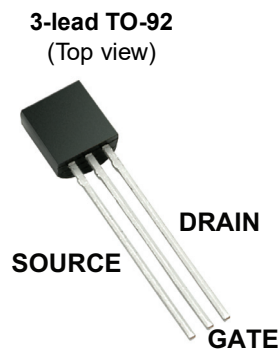
- Motor Controls
- Converters
- Amplifiers
- Switches
- Power Supply Circuits
- Drivers (Relays, Hammers, Solenoids, Lamps, Memory, Displays, Bipolar Transistors, etc.)

General Description

The 2N7000 is an Enhancement-mode (normally-off) transistor that uses a vertical DMOS structure and a well-proven silicon gate manufacturing process. This combination produces a device with the power handling capabilities of bipolar transistors as well as the high input impedance and positive temperature coefficient inherent in MOS devices. Characteristic of all MOS structures, this device is free from thermal runaway and thermally induced secondary breakdown.

Microchip's vertical DMOS FETs are ideally suited to a wide range of switching and amplifying applications where very low threshold voltage, high breakdown voltage, high input impedance, low input capacitance, and fast switching speeds are desired.

Package Type



See [Table 3-1](#) for pin information.

1.0 ELECTRICAL CHARACTERISTICS

Absolute Maximum Ratings†

Drain-to-Source Voltage	BV_{DSS}
Drain-to-Gate Voltage	BV_{DGS}
Gate-to-Source Voltage	$\pm 30V$
Operating Ambient Temperature, T_A	$-55^{\circ}C$ to $+150^{\circ}C$
Storage Temperature, T_S	$-55^{\circ}C$ to $+150^{\circ}C$

† **Notice:** Stresses above those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at those or any other conditions above those indicated in the operational sections of this specification is not intended. Exposure to maximum rating conditions for extended periods may affect device reliability.

DC ELECTRICAL CHARACTERISTICS

Electrical Specifications: $T_A = 25^{\circ}C$ unless otherwise specified. All DC parameters are 100% tested at $25^{\circ}C$ unless otherwise stated. (Pulse test: 300 μs pulse, 2% duty cycle)

Parameter	Sym.	Min.	Typ.	Max.	Unit	Conditions
Drain-to-Source Breakdown Voltage	BV_{DSS}	60	—	—	V	$V_{GS} = 0V, I_D = 10 \mu A$
Gate Threshold Voltage	$V_{GS(th)}$	0.8	—	3	V	$V_{GS} = V_{DS}, I_D = 1 mA$
Gate Body Leakage Current	I_{GSS}	—	—	10	nA	$V_{GS} = \pm 15V, V_{DS} = 0V$
Zero-Gate Voltage Drain Current	I_{DSS}	—	—	1	μA	$V_{GS} = 0V, V_{DS} = 48V$
		—	—	1	mA	$V_{GS} = 0V, V_{DS} = 48V, T_A = 125^{\circ}C$ (Note 1)
On-State Drain Current	$I_{D(ON)}$	75	—	—	mA	$V_{GS} = 4.5V, V_{DS} = 10V$
Static Drain-to-Source On-State Resistance	$R_{DS(ON)}$	—	—	5.3	Ω	$V_{GS} = 4.5V, I_D = 75 mA$
		—	—	5	Ω	$V_{GS} = 10V, I_D = 500 mA$

Note 1: Specification is obtained by characterization and is not 100% tested.

AC ELECTRICAL CHARACTERISTICS

Electrical Specifications: $T_A = 25^{\circ}C$ unless otherwise specified. All AC parameters are not 100% sample tested.

Parameter	Sym.	Min.	Typ.	Max.	Unit	Conditions
Forward Transconductance	G_{FS}	100	—	—	mmho	$V_{DS} = 10V, I_D = 200 mA$
Input Capacitance	C_{ISS}	—	—	60	pF	$V_{GS} = 0V, V_{DS} = 25V,$ $f = 1 MHz$
Common-Source Output Capacitance	C_{OSS}	—	—	25	pF	
Reverse Transfer Capacitance	C_{RSS}	—	—	5	pF	
Turn-On Delay Time	$t_{d(ON)}$	—	—	10	ns	$V_{DD} = 15V, I_D = 500 mA,$ $R_{GEN} = 25\Omega$
Turn-Off Delay Time	$t_{d(OFF)}$	—	—	10	ns	

DIODE PARAMETER

Diode Forward Voltage Drop	V_{SD}	—	0.85	—	V	$V_{GS} = 0V, I_{SD} = 200 mA$ (Note 1)
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Note 1: All DC parameters are 100% tested at $25^{\circ}C$ unless otherwise stated.
(Pulse test: 300 μs pulse, 2% duty cycle)

TEMPERATURE SPECIFICATIONS

Electrical Characteristics: Unless otherwise specified, for all specifications $T_A = T_J = +25^\circ\text{C}$.						
Parameter	Sym.	Min.	Typ.	Max.	Unit	Conditions
TEMPERATURE RANGE						
Operating Ambient Temperature	T_A	-55	—	+150	$^\circ\text{C}$	
Storage Temperature	T_S	-55	—	+150	$^\circ\text{C}$	
PACKAGE THERMAL RESISTANCE						
3-lead TO-92	θ_{JA}	—	132	—	$^\circ\text{C/W}$	

Note 1: Mounted on an FR4 board; 25 mm x 25 mm x 1.57 mm

THERMAL CHARACTERISTICS

Package	I_D (Note 1) (Continuous) (mA)	I_D (Pulsed) (mA)	Power Dissipation at $T_C = 25^\circ\text{C}$ (W)	I_{DR} (Note 1) (mA)	I_{DRM} (mA)
3-lead TO-92	200	500	1	200	500

Note 1: I_D (continuous) is limited by maximum T_J .

2.0 TYPICAL PERFORMANCE CURVES

Note: The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (e.g. outside specified power supply range) and therefore outside the warranted range.

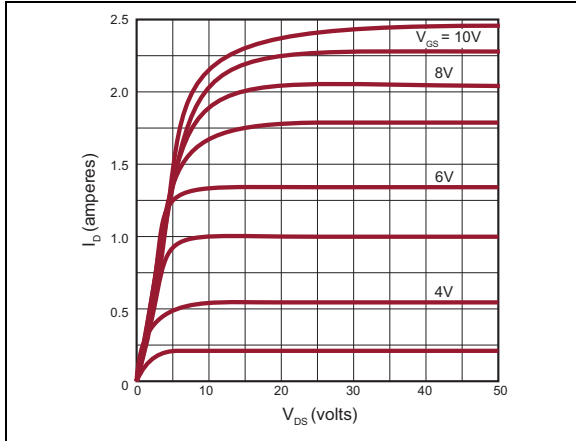


FIGURE 2-1: Output Characteristics.

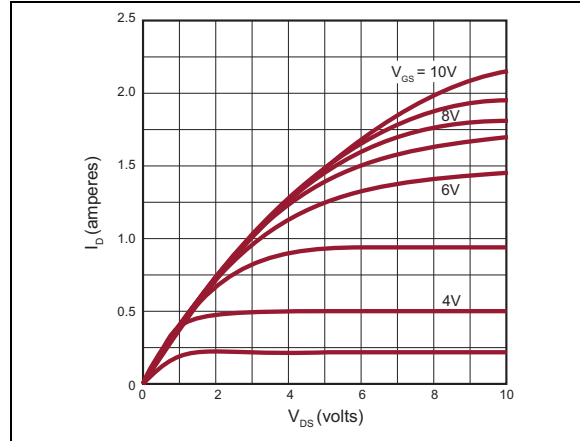


FIGURE 2-4: Saturation Characteristics.

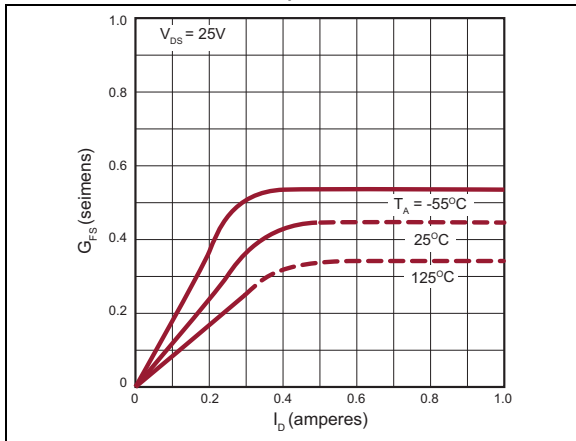


FIGURE 2-2: Transconductance vs. Drain Current.

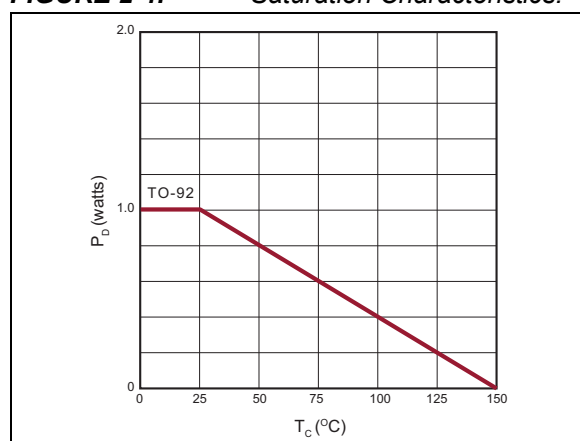


FIGURE 2-5: Power Dissipation vs. Case Temperature.

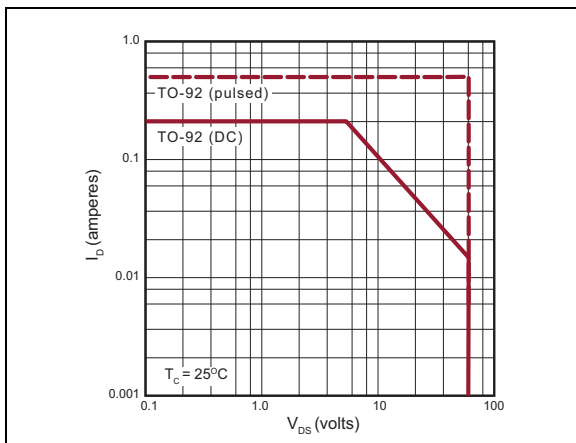


FIGURE 2-3: Maximum Rated Safe Operating Area.

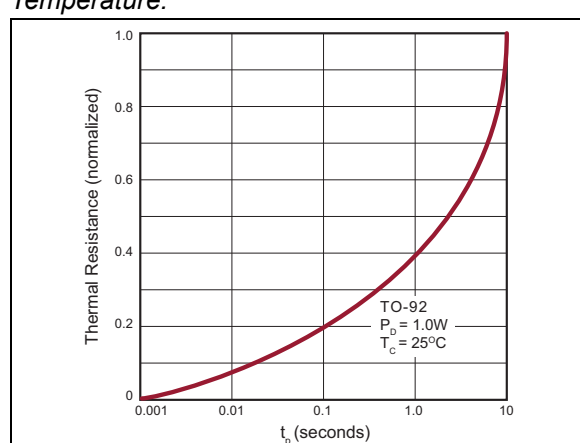


FIGURE 2-6: Thermal Response Characteristics.

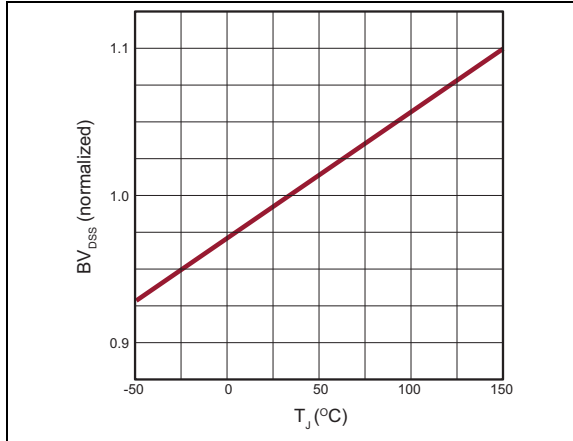


FIGURE 2-7: BV_{DSS} Variation with Temperature.

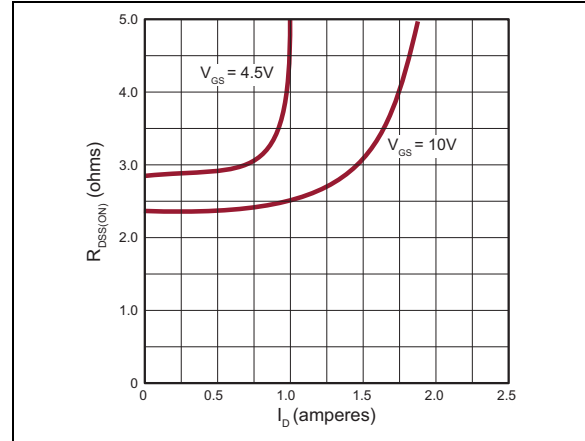


FIGURE 2-10: On-Resistance vs. Drain Current.

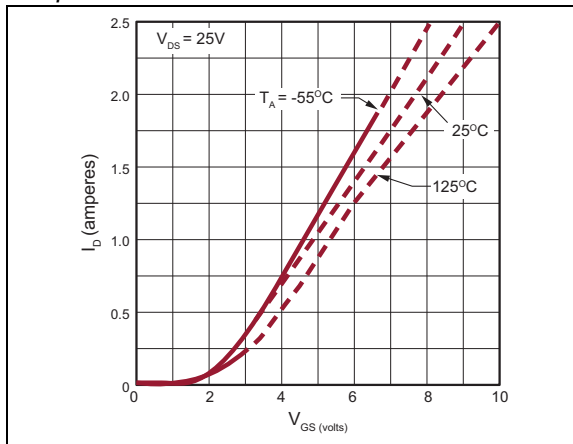


FIGURE 2-8: Transfer Characteristics.

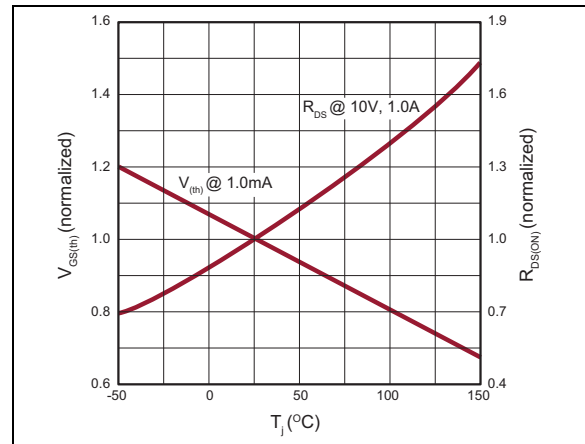


FIGURE 2-11: $V_{GS(th)}$ and R_{DS} Variation with Temperature.

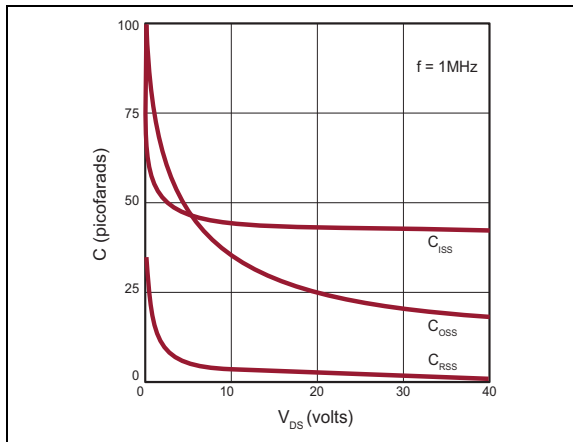


FIGURE 2-9: Capacitance vs. Drain-to-Source Voltage.

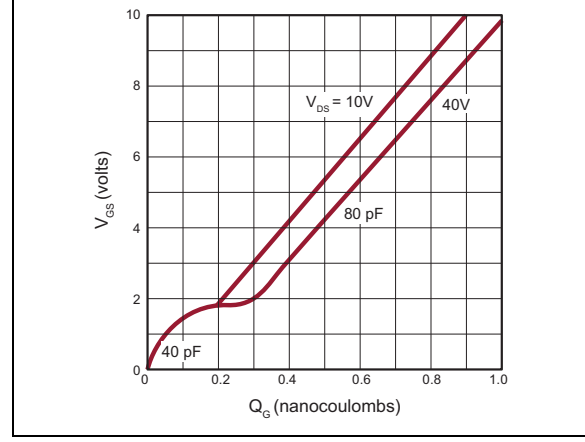


FIGURE 2-12: Gate Drive Dynamic Characteristics.